

# Formulaire d'intégration du champ de déformation

$\mathbb{E}_1$

$\mathbb{E}_2$

Champ de déformation initial

|                                         |          |          |
|-----------------------------------------|----------|----------|
| $\varepsilon_{11} =$                    | $ax_1^2$ | $0$      |
| $\varepsilon_{22} =$                    | $0$      | $ax_1^2$ |
| $\varepsilon_{33} =$                    | $0$      | $0$      |
| $\varepsilon_{12} = \varepsilon_{21} =$ | $0$      | $0$      |
| $\varepsilon_{23} = \varepsilon_{32} =$ | $0$      | $0$      |
| $\varepsilon_{31} = \varepsilon_{13} =$ | $0$      | $0$      |

Équations de compatibilité :  $\text{rot}(\text{rot } \underline{\underline{\varepsilon}}) = \underline{\underline{0}}$

$$\omega_{ij,kl} = \omega_{ij,lk} \Leftrightarrow \varepsilon_{ik,jl} - \varepsilon_{jk,il} = \varepsilon_{il,jk} - \varepsilon_{jl,ik}$$

Tenseur et vecteur de rotation

$$\underline{\underline{\omega}} = \begin{pmatrix} 0 & \omega_{12} & -\omega_{31} \\ -\omega_{12} & 0 & \omega_{23} \\ \omega_{31} & -\omega_{23} & 0 \end{pmatrix} = \begin{pmatrix} -\omega_{23} \\ -\omega_{31} \\ -\omega_{12} \end{pmatrix} \wedge$$

Vecteurs translation  $\vec{\lambda}$  et rotation  $\vec{\Omega}$  de corps rigide

$$\vec{\lambda} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} \quad \text{et} \quad \vec{\Omega} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

1. Calcul de  $\omega_{ij,k} = \varepsilon_{ki,j} - \varepsilon_{jk,i}$

|                                                             |     |          |
|-------------------------------------------------------------|-----|----------|
| $\omega_{12,1} = \varepsilon_{11,2} - \varepsilon_{21,1} =$ | $0$ | $0$      |
| $\omega_{12,2} = \varepsilon_{21,2} - \varepsilon_{22,1} =$ | $0$ | $-2ax_1$ |
| $\omega_{12,3} = \varepsilon_{31,2} - \varepsilon_{23,1} =$ | $0$ | $0$      |

|                                                             |     |     |
|-------------------------------------------------------------|-----|-----|
| $\omega_{23,1} = \varepsilon_{12,3} - \varepsilon_{31,2} =$ | $0$ | $0$ |
| $\omega_{23,2} = \varepsilon_{22,3} - \varepsilon_{32,2} =$ | $0$ | $0$ |
| $\omega_{23,3} = \varepsilon_{32,3} - \varepsilon_{33,2} =$ | $0$ | $0$ |

|                                                             |     |     |
|-------------------------------------------------------------|-----|-----|
| $\omega_{31,1} = \varepsilon_{13,1} - \varepsilon_{11,3} =$ | $0$ | $0$ |
| $\omega_{31,2} = \varepsilon_{23,1} - \varepsilon_{12,3} =$ | $0$ | $0$ |
| $\omega_{31,3} = \varepsilon_{33,1} - \varepsilon_{13,3} =$ | $0$ | $0$ |

3. Calcul de  $u_{i,j} = \varepsilon_{ij} + \omega_{ij}$

|                                              |          |             |
|----------------------------------------------|----------|-------------|
| $u_{1,1} = \varepsilon_{11} =$               | $ax_1^2$ | $0$         |
| $u_{1,2} = \varepsilon_{12} + \omega_{12} =$ | $-r +$   | $-2ax_1x_2$ |
| $u_{1,3} = \varepsilon_{13} - \omega_{31} =$ | $+q +$   | $0$         |

|                                              |        |            |
|----------------------------------------------|--------|------------|
| $u_{2,1} = \varepsilon_{21} - \omega_{12} =$ | $+r +$ | $2ax_1x_2$ |
| $u_{2,2} = \varepsilon_{22} =$               | $0$    | $ax_1^2$   |
| $u_{2,3} = \varepsilon_{23} + \omega_{23} =$ | $-p +$ | $0$        |

|                                              |        |     |
|----------------------------------------------|--------|-----|
| $u_{3,1} = \varepsilon_{31} + \omega_{31} =$ | $-q +$ | $0$ |
| $u_{3,2} = \varepsilon_{32} - \omega_{23} =$ | $+p +$ | $0$ |
| $u_{3,3} = \varepsilon_{33} =$               | $0$    | $0$ |

2. Calcul de  $\omega_{ij}$  par intégration de  $\omega_{ij,k}$

|                 |        |             |
|-----------------|--------|-------------|
| $\omega_{12} =$ | $-r +$ | $-2ax_1x_2$ |
| $\omega_{23} =$ | $-p +$ | $0$         |
| $\omega_{31} =$ | $-q +$ | $0$         |

4. Calcul de  $u_i$  par intégration de  $u_{i,j}$

|         |                             |                                |
|---------|-----------------------------|--------------------------------|
| $u_1 =$ | $\lambda_1 - rx_2 + qx_3 +$ | $\frac{a}{3}x_1^3 - ax_1x_2^2$ |
| $u_2 =$ | $\lambda_2 - px_3 + rx_1 +$ | $ax_1^2x_2$                    |
| $u_3 =$ | $\lambda_3 - qx_1 + px_2 +$ | $0$                            |

5. Vérification de la compatibilité de la déformation  $\hat{\varepsilon}_{ij}$  (calculée à partir de  $u_i$  à l'étape 4)  $\stackrel{?}{=} \varepsilon_{ij}$

|                                      |          |           |
|--------------------------------------|----------|-----------|
| $\hat{\varepsilon}_{11} = u_{1,1} =$ | $ax_1^2$ | $-ax_2^2$ |
| $\hat{\varepsilon}_{22} = u_{2,2} =$ | $0$      | $ax_1^2$  |
| $\hat{\varepsilon}_{33} = u_{3,3} =$ | $0$      | $0$       |

|                                                             |     |     |
|-------------------------------------------------------------|-----|-----|
| $\hat{\varepsilon}_{12} = \frac{1}{2}(u_{1,2} + u_{2,1}) =$ | $0$ | $0$ |
| $\hat{\varepsilon}_{13} = \frac{1}{2}(u_{1,3} + u_{3,1}) =$ | $0$ | $0$ |
| $\hat{\varepsilon}_{23} = \frac{1}{2}(u_{2,3} + u_{3,2}) =$ | $0$ | $0$ |

$\mathbb{E}_1 = \hat{\mathbb{E}}_1 \Rightarrow \mathbb{E}_1 \text{ compatible}$

$\mathbb{E}_2 \neq \hat{\mathbb{E}}_2 \Rightarrow \mathbb{E}_2 \text{ non compatible !}$